Aerosols and Clouds interactions

Computational Methods and tools

Emma Willaume
Tamara Maeder

Course by:
Dr. Satoshi Takahama
ENG-270

Fall Semester 2025

Contents

1 Deviation from Proposal

During the project, we made several adjustments to ensure accurate results and meet time constraints. We modified the method for calculating optical depth in the albedo function, as the original formula produced unrealistic values, and the adapted approach yielded accurate results. The planned analysis of a real case study was dropped due to time constraints and lack of a suitable example, so we focused solely on simulations for the selected aerosol sample.
While we initially intended to use HelloCloud as a data source and code reference, we relied entirely on Pyrcel and Particula, which provided sufficient data and results.
Despite these changes, the methodology and main objective of analyzing how aerosol microphysical properties affect cloud droplet activation and cloud albedo, remained unchanged.

2 Introduction to the problem

Aerosols, small particles suspended in the atmosphere, play a crucial yet highly uncertain role in regulating Earth's climate. They affect the climate system in two primary ways: directly, by scattering or absorbing incoming solar radiation, and indirectly, by modifying cloud characteristics through aerosol–cloud interactions (ACI) [1]. Among these indirect mechanisms, the Twomey effect is particularly significant: when aerosols serve as cloud condensation nuclei (CCN), they increase cloud droplet number concentration, thereby enhancing cloud reflectivity (albedo) for a given liquid water path [1, 2]. This mechanism alters the planet's radiation balance and contributes to climate forcing.

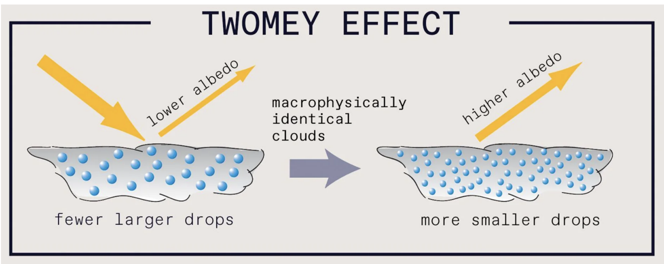

Figure 1: Twomey effect

Since preindustrial times, anthropogenic emissions have substantially intensified these effects. The study from IPCC [3] highlights various concentrations for each aerosols C.2 and properties of clouds C.1 across the world. We decided to focus on core physical processes and properties of aerosols (size, hygroscopicity, concentration) that influence cloud droplet activation and, in turn, cloud albedo through the Twomey effect, to understand why the emission of anthropogenic aerosols modify cloud albedo through the Twomey effect. However, we didn't considered the life time of aerosols in the atmosphere for this project.

3 Approach used

We used Python for the simulations because it provides specialized libraries such as Pyrcel and Particula, which simplify aerosol microphysics modeling through built-in functions for lognormal size distributions, CCN activation, and height–radius distributions. Python also enabled efficient visualization of the results. For the albedo calculations, we used C due to the need for repeated, large-scale summations over aerosol bins to compute optical depth across multiple scenarios. C was chosen for its superior performance in handling these computationally intensive operations.

3.1 Aerosols selected for each scenario

We decided to focus on five aerosols that are often present in the atmosphere. The parameters linked to each of them are summarized in table 1. N_0 is the total number concentration, D_g the geometric mean dry diameter, σ_g the geometric standard deviation, and κ the hygroscopicity parameter. In κ–Köhler theory, hygroscopicity controls how easily a particle can take up water once supersaturation is available. The values are taken from various articles: [4] and [5].

Aerosol mode	Origin	Light...	N_0 (cm^{-3})	D_g (μm)	σ_g	κ
Sea salt (coarse)	Natural	Scattering	10	1.7	2.1	1.20
Biogenic organic carbon	Natural	Scattering	300	0.15	1.6	0.15
Sulfate (accumulation)	Anthropogenic	Scattering	850	0.03	1.7	0.54
Anthropogenic OC	Anthropogenic	Scattering	400	0.12	1.6	0.20
Black carbon (fine)	Anthropogenic	Absorbing	200	0.05	1.5	0.01

Table 1: Micro-physical parameters of the aerosol modes used in the Pyrcel simulations.

3.2 Lognormal distribution

To model the composition of our volume of the atmosphere before the simulations, we used two models to calculate and plot the lognormal size distribution of our particles. The comparison enabled us to understand which model was the most appropriate to do the rest of our simulations and understand if the results obtained are coherent and reliable.

There are many ways to represent aerosol size distribution. We focus on the lognormal size distribution because it is experimentally often the case[6]. Thus it represents well the complexity of the distribution of sizes. In addition, we decided to focus on couple of aerosol with a multi-mode approach that correspond more to the reality where natural and anthropogenic aerosols are mixed. The formula of the lognormal distribution is the following [7]:

$$\frac{dN}{d\log d} = \sum_{i=1}^{N} \frac{N_i}{\log\sigma\sqrt{2\pi}} e^{-\dfrac{(\log d - \log d_i)^2}{2\log^2\sigma}} \tag{1}$$

Where d$_i$, σ and N$_i$ represent respectively, the geometric diameter, the standard deviation and the concentration of particles of this mode.

We used the library Pyrcel [8] to calculate the distribution density function dN/dlog(x) (more information upon this function in Appendix B).

Then, the documentation of Particula highlighted the fact that the probability mass function (PMF) was more convenient than probability density function (PDF), to proceed air quality monitoring where knowing the exact count of particles at different sizes is more necessary. Indeed, while the PDF represents the probability of particle sizes occurring within a continuous range, the PMF deals directly with discrete particle counts in each size bins [9]. As a consequence, we decided to use Particula[10] do calculate it and compare both (more information upon the function used in Appendix B).

3.3 Droplet Activation

The air is supersaturated when the saturation goes above 100%. The air contains more water vapor that it can at a certain temperature. As the particles go up in the atmosphere, the temperature decreases and the supersaturation increases. At a certain point, the supersaturation is too high and the water condense around aerosols which leads to a rapid growth of the droplet radius for particles with a radius sufficiently important. Then, the supersaturation decreases because the activation of droplets reduces the amount of water in the air. We used a specific function from Pyrcel (more information in Appendix B) to track the growth of droplets in each bins and at every height.

3.4 Optical depth model

Our initial approach to compute the optical depth τ was based on the effective radius and the mean volumetric radius of cloud droplets. However, this method led to unrealistic values due to difficulties in obtaining data with the correct magnitude. To address this issue, we instead relied on bin-resolved aerosol information, namely the wet radius and number concentration of activated particles.

Because we considered a mixture of two aerosol species, the cloud optical depth was computed by summing the contributions of all size bins from both aerosols. The optical depth was thus calculated as

$$\tau = 2\pi \left(\sum_i r_{i,1}^2 N_{i,1} + \sum_j r_{j,2}^2 N_{j,2} \right),$$

where $r_{i,k}$ and $N_{i,k}$ are respectively the mean wet radius (in meters) and the number concentration (in m^{-3}) of bin i for aerosol species k. This formulation accounts for the combined extinction contributions of both aerosol populations to the cloud depth.

The resulting optical depth τ was then used in the albedo parameterization described in Section 3.5, yielding the cloud albedo and allowing us to assess the impact of mixed aerosol populations on cloud reflectivity.[11]

3.5 Albedo model

To compute the cloud albedo for each binary aerosol mixing, we used the formulation proposed by Dupuy (2003), which expresses the albedo as

$$R = \frac{\tau}{\left(\frac{a}{1-g}\right) + \tau},$$

where R denotes the cloud albedo, τ is the optical depth (previously calculated in Section 3.4), $a = 2$ is a constant, and $g = 0.85$ is the asymmetry parameter. This formulation relates the cloud's scattering properties directly to its reflectivity.

For the implementation, we first developed a C function that computes the albedo from a given optical depth. This function was then called by a separate program to process all aerosol data contained in multiple CSV files. For each mixing scenario, the program applied the formulation and computed the corresponding cloud albedo. These results form the basis for comparing the impact of different aerosol mixtures on cloud albedo, with a particular focus on the role of anthropogenic aerosols, which is the primary objective of this analysis.[11]

4 Results

4.1 Lognormal distribution from both Particula and Pyrcel model

We plotted the lognormal size distribution resulting from PDF for each couple of aerosols with Pyrcel (graphs without grid patterns) and then the one from Particula (with grid patterns) resulting from PMF calculations. Only the graph for the scenario with sulfate and sea salt is shown here but we will also discuss about the other scenarios. The additional graphs and analyzes are in appendix A.

On a qualitative point of view, we can observe that in each comparison cases, the particles radius are following the same trends and overlap for similar values which confirms the credibility of both models. Quantitatively, in the figure 2 we can observe a maximum number concentration around 17.5 cm^3 and a maximum number of sulfate particles around 3. It is a good example of the importance of the PMF approach to get more easily an idea of the quantities in each bins.

Concerning the size distribution of each multi-mode calculated by Pyrcel. We can observe in figure 2 that the sulfate has a greater concentration than sea salt but that its dry radius is one hundred times smaller so both distributions do not overlap.

It is quite clear that the larger radius is the one of sea salt particles but they are also the least present in the atmosphere. Thus, it is important to note that anthropogenic aerosols (sulfate, anthropogenic organic compounds and black carbon) have a peak concentration for radius under 0.1 μm but are reaching higher concentrations than natural aerosols.

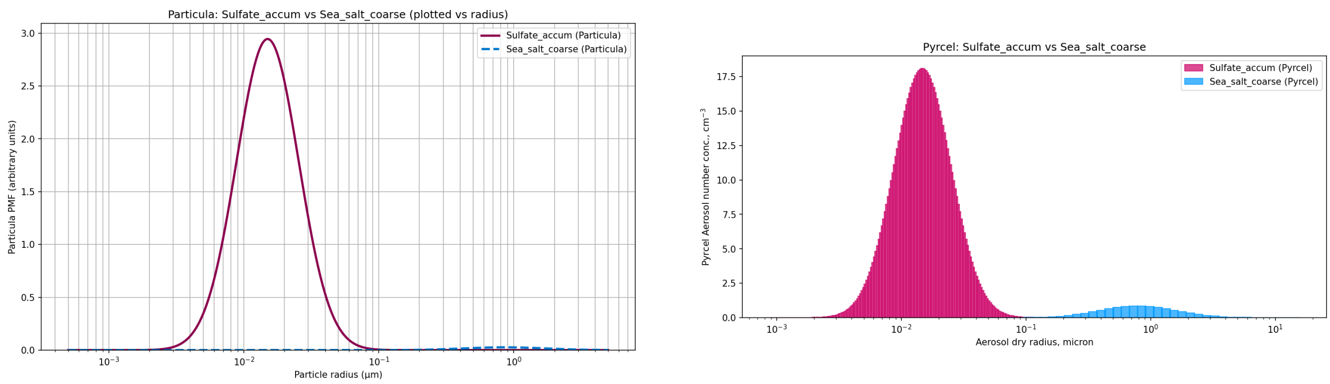

Figure 2: Lognormal distribution for a simulation with sea salt and sulfate.
Particula on the left and Pyrcel on the right.

4.2 Simulations of droplet growth

The figures 3 and in appendix A figures: A.7, A.9 and A.8 are representing on the right the height vs the supersaturation on the bottom axis and the temperature on the top axis. The maximum supersaturation is indicated with an arrow. The curve of the height vs the temperature is the same for each case and it is used only to indicate that the supersaturation depends on temperature. Three general trends about the growth of droplet radius are observable in each scenarios. First, only the dry radius above a certain value, form droplets. Under this threshold, the radius are constant

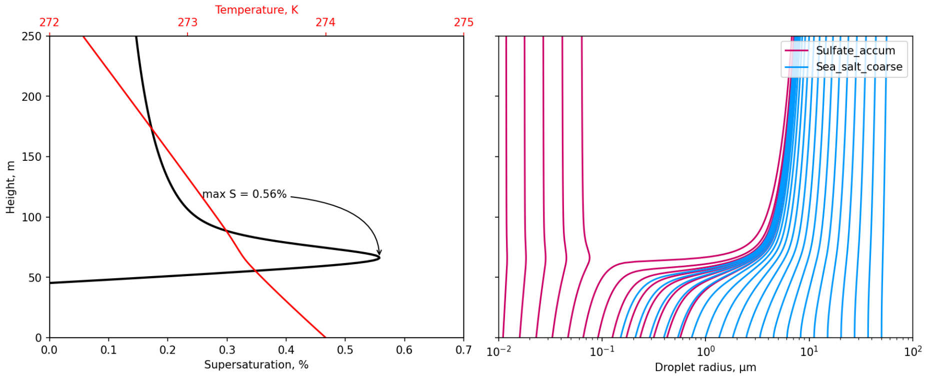

Figure 3: Simulation from Pyrcel with sea salt and sulfate, a)Height vs supersaturation and temperature, b) height vs droplet radius

at every height because no water condenses around the aerosol. Secondly, the growth of droplet occurs at a certain height that correspond to the maximum saturation that is reached in the cloud. The third phase is a stabilization of the droplet radius for the entire cloud (i.e. at all heights).

Let us analyze each scenario to compare the values. The table 2 represents the key parameters of each cases. To analyze those values, the parameters related to each aerosols summarized in table 1 will be use.

Parameter	**Figure 7**	**Figure A.9**	**Figure A.8**	**Figure A.7**
Aerosol 1	Sea salt	Biogenic OC	Sea salt	Sea salt
Aerosol 2	Sulfate	Anthropogenic OC	Black carbon	Biogenic OC
Maximum supersaturation (%)	0.56	0.42	–	0.40
Minimum dry radius to grow (μm)	0.05	0.08	0.05	0.08
Minimal droplet radius at 250 m(μm)	0.5	0.3	1	0.5

Table 2: Summary of the key parameters of each scenario

In figure 3, the sulfate that has a smaller dry radius is only partly activated while all the sea salt is activated. The supersaturation in this case is the higher value obtained between all scenarios. A reason could be that the number concentration of sulfate particles is the highest between our scenario which means that a lot of particles compete for water vapor. As a result, supersaturation builds up more slowly and reaches a higher peak before dropping. Another observation to point out is that some sea salt particles do not grow more (at least not visible on a log scale). This happens because their dry radius is so large that the critical supersaturation is very low and they are immediately activated. Further analysis for additional plots are given in Appendix A.

4.3 Albedo

The results presented in figure 4 highlight a strong dependence of cloud albedo on aerosol composition. First, the presence of absorbing aerosols such as black carbon leads to a significant reduction in cloud albedo compared to simulations with scattering aerosols. Then, scenarios dominated by organic aerosols exhibit the highest albedo due to efficient CCN activation and enhanced droplet number concentrations, consistent with a strong Twomey effect. In contrast, although sulfate aerosols are highly hygroscopic, their combination with coarse sea-salt particles leads to competition for available water vapor during activation and growth. This competition limits the total number of activated droplets and favors the growth of fewer, larger droplets, resulting in a lower cloud albedo compared to the organic-dominated scenario. Competition is also the reason why in figure 5, the effective radius and albedo decrease with number concentration of sulfate.

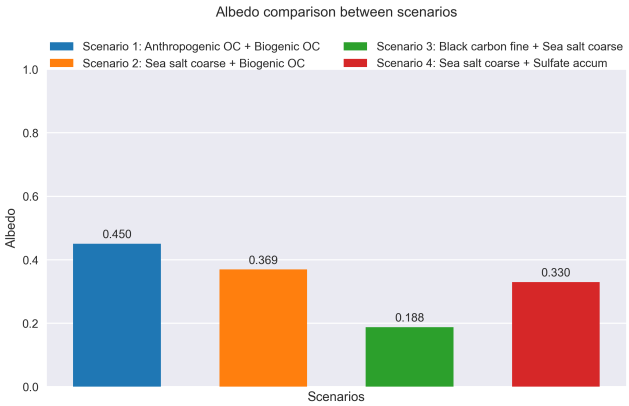

Figure 4: Albedo comparison between scenarios

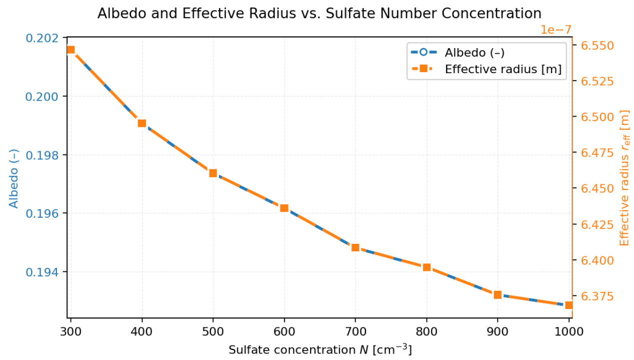

Figure 5: Albedo and Effective radius vs number concentration

5 Conclusion and Outlook

To conclude, our model presents several limitations. First, it does not account for the lifetime effect. Consequently, the interaction between competition processes and droplet growth, particularly for sulfate and sea salt aerosols, complicates an accurate assessment of the impact of increasing anthropogenic aerosol number concentrations. Nevertheless, the model successfully highlights the influence of anthropogenic aerosols on cloud albedo, demonstrating that clouds exhibit higher albedo in the presence of scattering aerosols compared to the case with absorbing aerosols.

A valuable direction for future work would be to investigate the impact of individual aerosol concentrations in the China C case study and to explain why optical depth is particularly high in certain regions. However, achieving a more realistic understanding would require more robust datasets and a better representation of aerosol lifetime effects and aerosols-cloud-interactions. Overall, this project emphasizes the importance of detailed mathematical modeling and the need for more accessible and comprehensive data to improve the accuracy and applicability of future studies. Despite its limitations, this work contributes to a better understanding of how anthropogenic aerosols influence cloud droplet activation and cloud reflectance.

6 Authorship Statement

The work was divided between code development and theoretical research. Emma conducted the research on the lognormal distribution and droplet activation, and developed the Python code using the Pyrcel and Particula libraries to compute aerosol profiles for each scenario and generate the corresponding plots. Tamara was in charge of the computation of albedo and effective radius using C, as well as for processing the datasets generated across scenarios and producing the associated visualizations. Emma also implemented the project automation by creating a shell script and a YAML configuration file, enabling reproducibility of the simulations, while Tamara completed the shell script and the documentation related to the C code. The report was jointly written by both authors. Overall, the workload was evenly distributed, and the collaboration was effective.

References

[1] A. Gettelman. "Putting the clouds back in aerosol–cloud interactions". In: *Atmospheric Chemistry and Physics* 15.21 (2015), pp. 12397–12411. DOI: 10.5194/acp-15-12397-2015.

[2] U. Lohmann and J. Feichter. "Can the direct and semi-direct aerosol effect compete with the indirect effect on a global scale?" In: *Geophysical Research Letters* 28.1 (2001), pp. 159–161. DOI: 10.1029/2000GL012051.

[3] O. Boucher et al. "Clouds and Aerosols". In: *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change.* Ed. by T.F. Stocker et al. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2013.

[4] P. Koepke M. Hess and I. Schult. "Optical Properties of Aerosols and Clouds: The Software Package OPAC". In: *Bulletin of the American Meteorological Society* (1998). URL: https://cds-espri.ipsl.upmc.fr/etherTypo/fileadmin/files/GEISA/hess-etal2.pdf.

[5] et S. M. Kreidenweis. Petters M. D. "A Single Parameter Representation of Hygroscopic Growth and Cloud Condensation Nucleus Activity". In: *Atmospheric Chemistry and Physics, Copernicus Online Journals* (2007). URL: https://doi.org/10.5194/acp-7-1961-2007.

[6] Observatoire Midi*Pyrénée*. *Physique des Aerosols*. URL: https://omer7a.obs-mip.fr/files/Physique_aerosols.pdf.

[7] Maxime HERVO. "Etude des propriétés optiques et radiatives des aérosols en atmosphère réelle: Impact de l'hygroscopicité." PhD thesis. Université Blaise Pascal – Clermont Ferrand, 2013.

[8] Daniel Rothenberg. *The simplest tools available for describing the growth and evolution of a cloud droplet spectrum from a given population of aerosols are based on zero-dimensional, adiabatic cloud parcel models.* URL: https://pyrcel.readthedocs.io/en/latest/sci_descr.html.

[9] Los Alamos National Laboratory Kyle Gorkowski. *Size ditributions tutorial for Particula*. URL: https://uncscode.github.io/particula/Examples/Particle_Phase/Notebooks/Aerosol_Distributions/.

[10] Los Alamos National Laboratory Kyle Gorkowski. *Particula is an open-source, Python-based aerosol simulator.* URL: https://uncscode.github.io/particula/.

[11] Régis Dupuy. "Étude de l'activation des noyaux de condensation : mesure, analyse et développement instrumental". Thèse de doctorat. Université Blaise Pascal – Clermont-Ferrand II, 2007. URL: https://wwwobs.univ-bpclermont.fr/atmos/fr/Theses/Th_Dupuy.pdf.

Appendix A

Additional graphs and analysis

Lognormal distribution analysis

Figure A.1 highlight the fact that black carbon particles have a low dry radius and a peak concentration around $11/cm^3$ while sea salt follow the same trend that we observed in figure 1 with sulfate.

Figure A.3 shows that organic compounds whether they are biogenic or anthropogenic have similar size distribution which was predictable because they have a similar chemical structure.
For the last case, represented on figure A.5, we choose a multi-mode with only natural aerosols but the concentration of biogenic organic compound is largely higher than the concentration of sea salt. The radius of organic compounds are between 0,05 μm and 0.5 μm with a peak around 0,1 μm while the peak of sea salt concentration occurs around $1 \mu m$.

Height vs radius analysis

In figure A.9, the aerosols coupled are very similar, as a consequence, they grow identically. It can be inferred that the number of droplet in the atmosphere will be almost doubled. Thus, we can observe in table 2 that the wet radius in the cloud are very small compared to other situations with sea salt.

The curve of the supersaturation in figure A.8 is uncommon. A reason could be that the hygroscopicity of the black carbon is very low. This means that it is also hydrophobic and it explains why it stay inactivated higher than sea salt. We can also conclude that the behavior of sea salt remains the same as for other scenarios which signifies that it is almost unaffected even if it is competing with various compounds.

As for figure A.7, the simulation was made to analyze a case without anthropogenic aerosols. However, the behavior of both sea salt and biogenic organic compounds are similar to the other scenario. The only difference is that the supersaturation is slightly lower than in other cases.

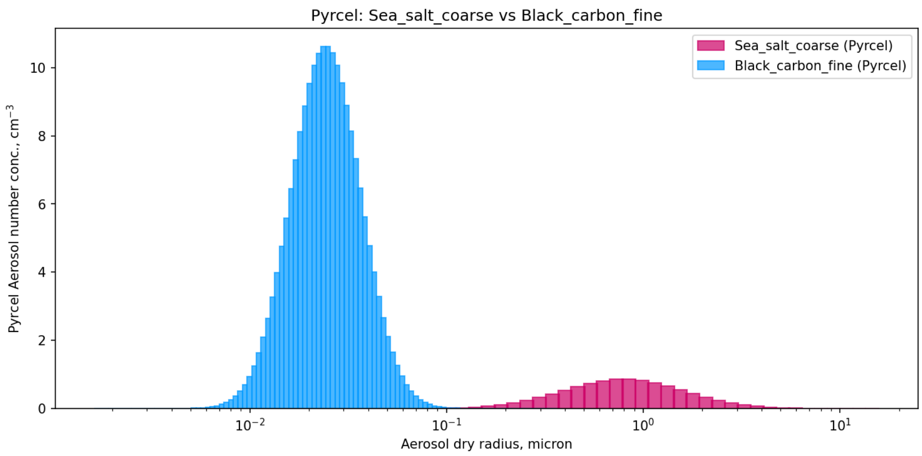

Figure A.1: Lognormal distribution made by Pyrcel for a simulation with sea salt and black carbon

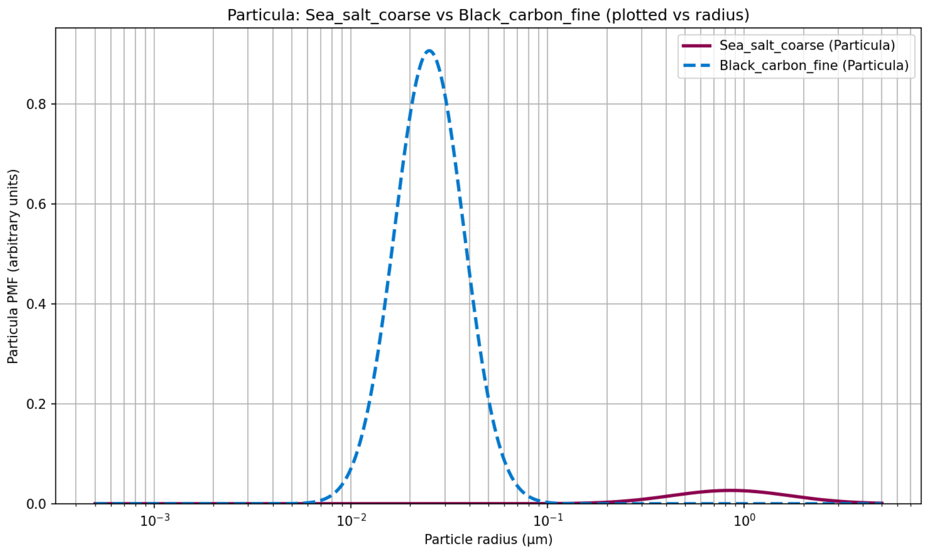

Figure A.2: Lognormal distribution made by Particula for a simulation with sea salt and black carbon

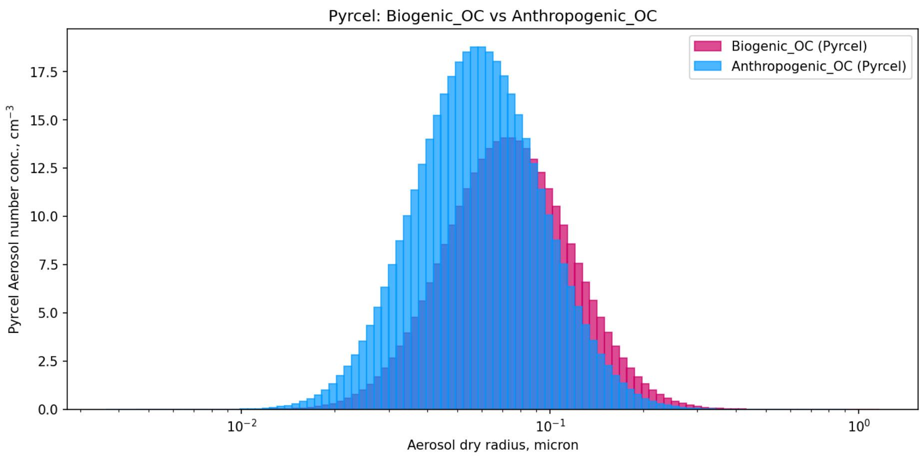

Figure A.3: Lognormal distribution made by Pyrcel for a simulation with biogenic and anthropogenic organic compounds

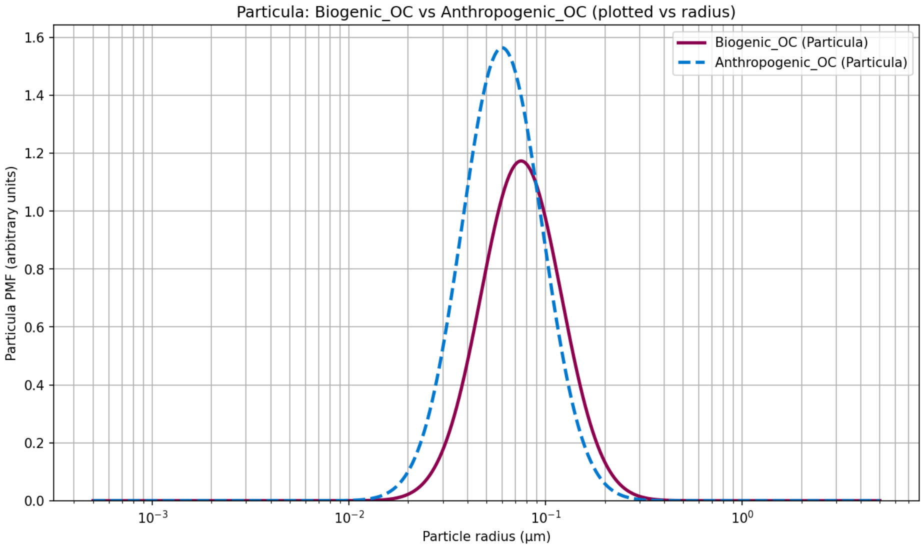

Figure A.4: Lognormal distribution made by Particula for a simulation with biogenic and organic compounds

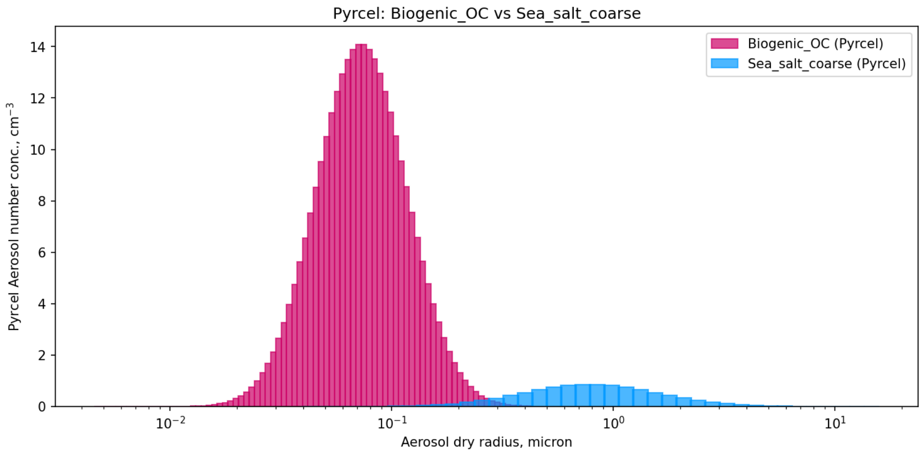

Figure A.5: Lognormal distribution made by Pyrcel for a simulation with sea salt and biogenic organic compounds

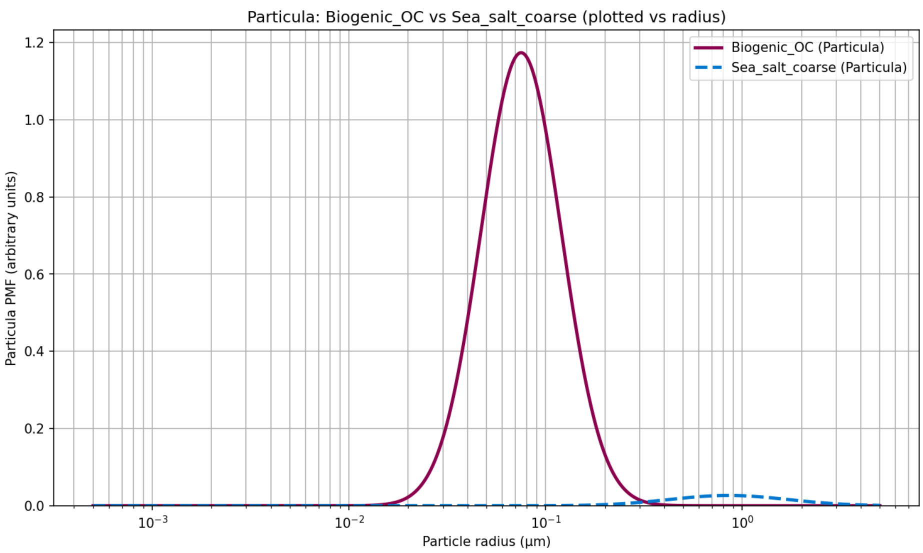

Figure A.6: Lognormal distribution made by Particula for a simulation with sea salt and biogenic organic compounds

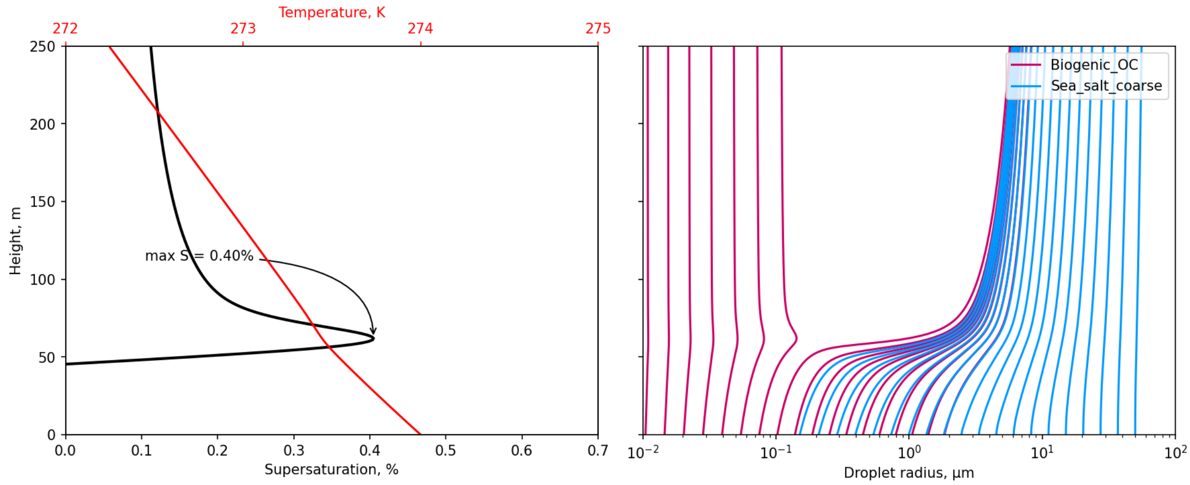

Figure A.7: Simulation from Pyrcel with sea salt and biogenic organic compounds
a)Height vs supersaturation and temperature, b) height vs droplet radius

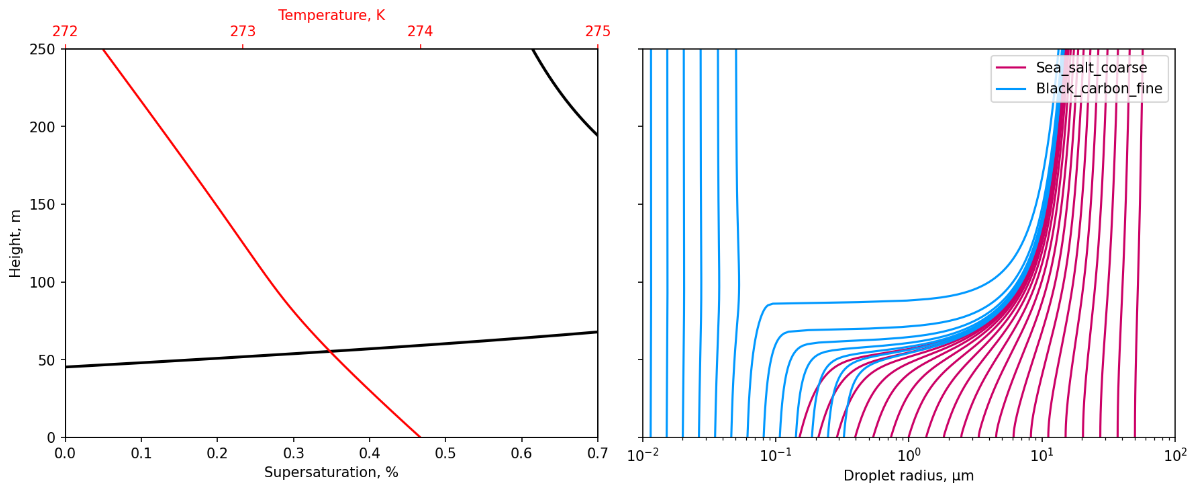

Figure A.8: Simulation from Pyrcel with sea salt and black carbon
a)Height vs supersaturation and temperature, b) height vs droplet radius

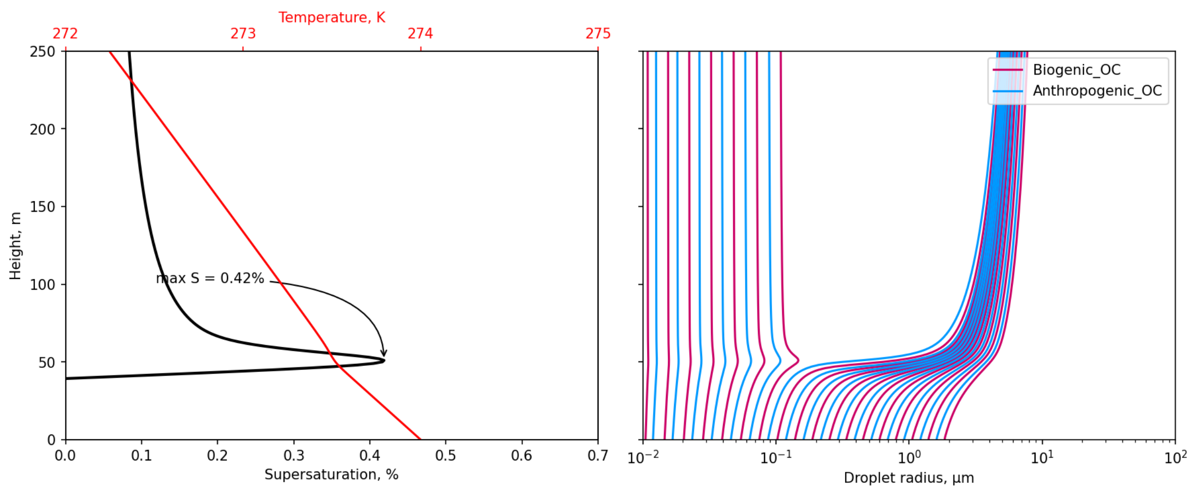

Figure A.9: Simulation from Pyrcel with biogenic and anthropogenic organic compounds
a)Height vs supersaturation and temperature, b) height vs droplet radius

Appendix B

Explanations of useful functions from each library

Particula
Function: get_log_normal_pmf_distribution
Aim: compute a lognormal probability mass function (PMF) for given modes.
Steps: This function first calculates the lognormal probability density function (PDF), then converts it to a PMF by integrating over x_values which are a range of particle diameters.

Arguments:

- x_values : 1D array of size points at which the PMF is evaluated.
- mode : Array of lognormal mode (scale) values for each mode.
- geometric_standard_deviation : Array of geometric standard deviation values for each mode.
- number_of_particles : Number of particles in each mode.

Outputs: 1D array of the total PMF values summed across all modes.

Pyrcel

Function: pdflog10.py
Aim: Calculate the distribution density function dN/dlog(x) (base 10 logarithm)
Arguments:

- mu : Radius in micrometers
- sigma : Standard deviation of the radius
- N : number concentration per cm^3

Function: Run from the class ParcelModel
Aim: Droplet growth
Arguments:

- all the parameters linked to an aerosol that are required to calculate the lognormal distribution and the lognormal distribution itself.

- kappa: the hygroscopicity.
- bins: the number of bins for the radius.
- P: Pressure: 77500 Pa.
- T0: Initial Temperature: 274 K.
- S0: Initial Supersaturation: 1-RH= -0.02 % .
- V: Updraft speed: 1 m/s.
- t_end: Total time over interval over which the model should be integrated: 250 s.
- dt: the time step for the integration: 1 s.
- all the other parameters are optional and we decided to keep the values from already implemented.

Outputs: two dataframes.
The first one contains profiles of the meteorological quantities tracked in the model, and the second a dictionary of dataframes with one for each aerosol species, tracking the growth in each bin for those species.

Appendix C

Case study of China

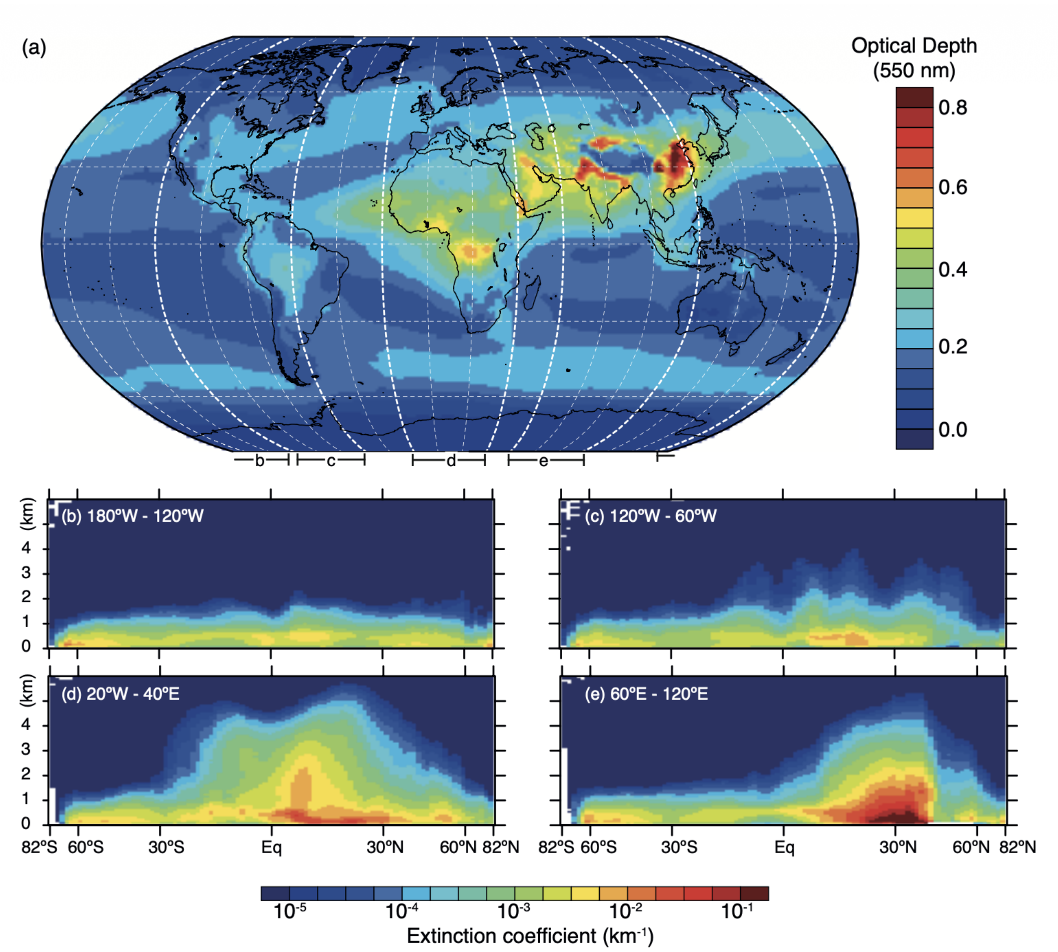

Figure 7.14 | (a) Spatial distribution of the 550 nm aerosol optical depth (AOD, unitless) from the European Centre for Medium Range Weather Forecasts (ECMWF) Integrated Forecast System model with assimilation of Moderate Resolution Imaging Spectrometer (MODIS) aerosol optical depth (Benedetti et al., 2009; Morcrette et al., 2009) averaged over the period 2003–2010; (b–e) latitudinal vertical cross sections of the 532 nm aerosol extinction coefficient (km^{-1}) for four longitudinal bands (180°W to 120°W, 120°W to 60°W, 20°W to 40°E, and 60°E to 120°E, respectively) from the Cloud–Aerosol Lidar with Orthogonal Polarization (CALIOP) instrument for the year 2010 (nighttime all-sky data, version 3; Winker et al., 2013).

Figure C.1: Spatial distribution of the 550 nm aerosol optical depth from the European Centre for Medium Range Weather Forecasts [3].

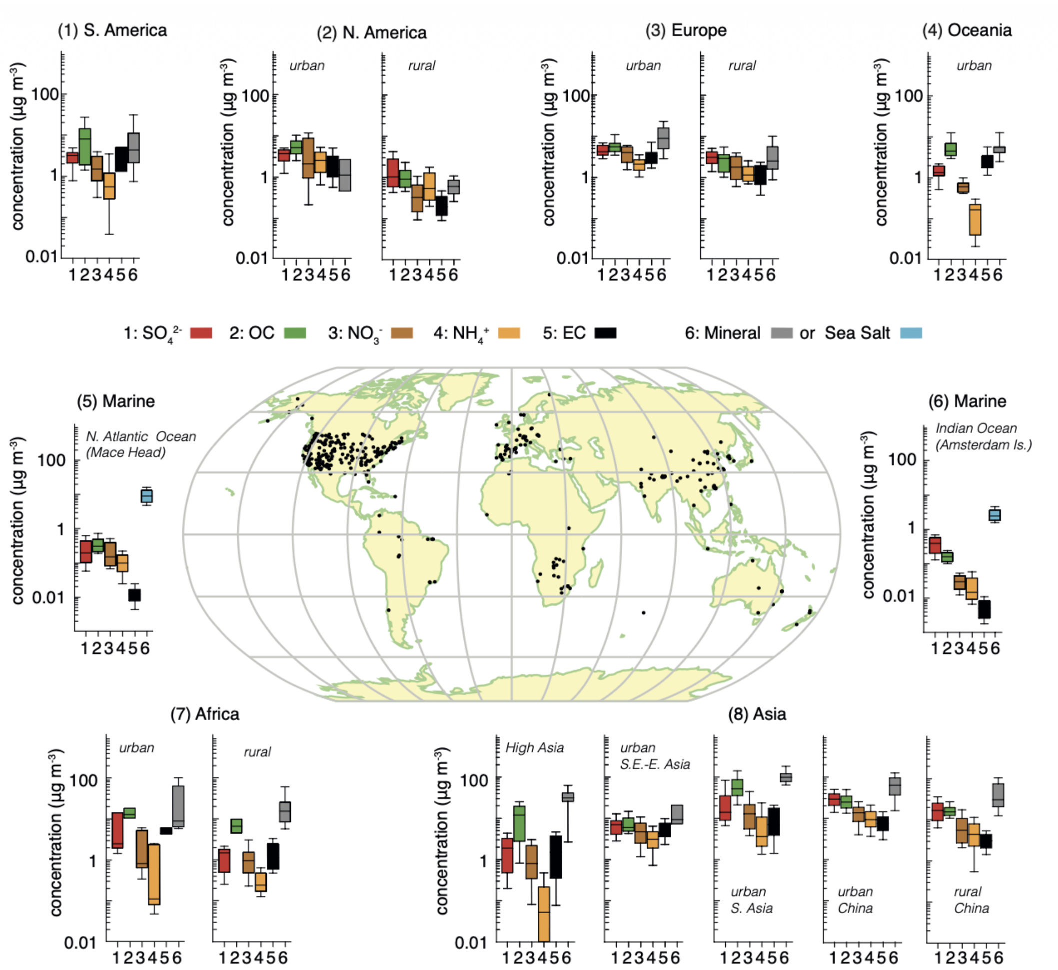

Figure 7.13 | Bar chart plots summarizing the mass concentration (µg m⁻³) of seven major aerosol components for particles with diameter smaller than 10 µm, from various rural and urban sites (dots on the central world map) in six continental areas of the world with at least an entire year of data and two marine sites. The density of the sites is a qualitative measure of the spatial representativeness of the values for each area. The North Atlantic and Indian Oceans panels correspond to measurements from single sites (Mace Head and Amsterdam Island, respectively) that are not necessarily representative. The relative abundances of different aerosol compounds are considered to reflect the relative importance of emissions of these compounds or their precursors, either anthropogenic or natural, in the different areas. For consistency the mass of organic aerosol (OA) has been converted to that of organic carbon (OC), according to a conversion factor (typically 1.4 to 1.6), as provided in each study. For each area, the panels represent the median, the 25th to 75th percentiles (box), and the 10th to 90th percentiles (whiskers) for each aerosol component. These include: **(1) South America** (Artaxo et al., 1998; Morales et al., 1998; Artaxo et al., 2002; Celis et al., 2004; Bourotte et al., 2007; Fuzzi et al., 2007; Mariani and Mello, 2007; de Souza et al., 2010; Martin et al., 2010a; Gioda et al., 2011); **(2) North America** with **urban United States** (Chow et al., 1993; Kim et al., 2000; Ito et al., 2004; Malm and Schichtel, 2004; Sawant et al., 2004; Liu et al., 2005); and **rural United States** (Chow et al., 1993; Malm et al., 1994; Malm and Schichtel, 2004; Liu et al., 2005); **(3) Europe** with **urban Europe** (Lenschow et al., 2001; Querol et al., 2001, 2004, 2006, 2008; Roosli et al., 2001; Rodriguez et al., 2002, 2004; Putaud et al., 2004; Hueglin et al., 2005; Lonati et al., 2005; Viana et al., 2006, 2007; Perez et al., 2008; Yin and Harrison, 2008; Lodhi et al., 2009); and **rural Europe** (Gullu et al., 2000; Querol et al., 2001, 2004, 2009; Rodriguez et al., 2002; Putaud et al., 2004; Puxbaum et al., 2004;Rodrıguez et al., 2004; Hueglin et al., 2005; Kocak et al., 2007; Salvador et al., 2007; Yttri, 2007; Viana et al., 2008; Yin and Harrison, 2008;Theodosi et al., 2010); **(4) urban Oceania** (Chan et al., 1997; Maenhaut et al., 2000; Wang and Shooter, 2001; Wang et al., 2005a; Radhi et al., 2010); **(5) marine northern Atlantic Ocean** (Rinaldi et al., 2009; Ovadnevaite et al., 2011); **(6) marine Indian Ocea**n (Sciare et al., 2009; Rinaldi et al., 2011); **(7) Africa** with **urban Africa** (Favez et al., 2008; Mkoma, 2008; Mkoma et al., 2009a); and **rural Africa** (Maenhaut et al., 1996; Nyanganyura et al., 2007; Mkoma, 2008, 2009a, 2009b; Weinstein et al., 2010); **(8) Asia** with **high Asia**, with altitude larger than 1680 m (Shresth et al., 2000; Zhang et al., 2001, 2008, 2012b; Carrico et al., 2003; Rastogi and Sarin, 2005; Ming et al., 2007a; Rengarajan et al., 2007; Qu et al., 2008; Decesari et al., 2010; Ram et al., 2010); **urban Southeast and East Asia** (Lee and Kang, 2001; Oanh et al., 2006; Kim et al., 2007; Han et al., 2008; Khan et al., 2010); **urban South Asia** (Rastogi and Sarin, 2005; Kumar et al., 2007; Lodhi et al., 2009; Chakraborty and Gupta, 2010; Khare and Baruah, 2010; Raman et al., 2010; Safai et al., 2010; Stone et al., 2010; Sahu et al., 2011); **urban China** (Cheng et al., 2000; Yao et al., 2002; Zhang et al., 2002; Wang et al., 2003, 2005b, 2006; Ye et al., 2003; Xiao and Liu, 2004; Hagler et al., 2006; Oanh et al., 2006; Zhang et al., 2011, 2012b); and **rural China** (Hu et al., 2002; Zhang et al., 2002; Hagler et al., 2006; Zhang et al., 2012b).

Figure C.2: Bar chart plots summarizing the mass concentration ($\mu g.m^{-3}$) of seven major aerosol components for particles with diameter smaller than 10 μm), from various rural and urban sites [3].